

Day 13 – Count Digits in a Number (Part 1)

Aaj ka topic:

Count Digits in a Number

Ye Basic Mathematics ka bahut important topic hai. Iska logic aage chal kar Reverse Number, Armstrong Number, Palindrome Number, Digit Sum, Digital Root, aur kai interview questions me use hoga.

Problem Statement

Ek integer n diya hai.

Usme total kitne digits hain, wo count karna hai.

Example 1

Input:
12345

Output:
5

Example 2

Input:
7

Output:
1

Example 3

Input:
908765

Output:
6

Kitni Approaches Hain?

Is problem ke liye mainly 3 **approaches** hain.

Approach 1 – Loop + Division ☐☐☐☐☐

- Standard Interview Approach
 - $O(\log N)$
 - $O(1)$
-

Approach 2 – String Method

- Easy
 - Interview me primary approach nahi
-

Approach 3 – Logarithm (Math)

- Fast Mathematical Formula
- Competitive Programming me useful
- Edge cases samajhna zaruri

Aaj **Part 1** me hum **Approach 1** padhenge.

Approach 1 – Division by 10

Main Idea

Har baar last digit hatao.

Jab number 0 ban jaye tab loop stop.

Jitni baar loop chala,

utni digits.

Example

12345

Remove last digit

1234

Again

123

Again

12

Again

1

Again

0

Kitni baar remove kiya?

5

Answer

5 digits

Logic

Har iteration me

```
n = n / 10;
```

Aur

```
count++;
```

Bas.

Visualization

Input

58291

Iteration 1

5829

Count

1

Iteration 2

582

Count

2

Iteration 3

58

Count

3

Iteration 4

5

Count

4

Iteration 5

0

Count

5

Done.

Java Code

```
import java.util.Scanner;

public class CountDigits {

    public static int countDigits(int n) {

        if (n == 0) {
            return 1;
        }

        n = Math.abs(n);

        int count = 0;

        while (n != 0) {

            count++;

            n = n / 10;
        }

        return count;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number : ");

        int n = sc.nextInt();

        System.out.println("Digits = " + countDigits(n));
    }
}
```

Line by Line Explanation

Line

```
if (n == 0)
```

Ye special case hai.

Agar input

0

To kitni digits?

1

Kyunki 0 khud ek digit hai.

Isliye

```
return 1;
```

Line

```
n = Math.abs(n);
```

Example

-12345

Ban jayega

12345

Minus sign digit nahi hota.

Hum sirf digits count karna chahte hain.

Line

```
int count = 0;
```

Digits count karega.

Line

```
while(n != 0)
```

Jab tak number khatam nahi hota.

Line

```
count++;
```

Ek digit remove hone wali hai.

Isliye count increase.

Line

```
n = n / 10;
```

Last digit hata do.

Example

9876

↓

987

Line

```
return count;
```

Final digits return.

Complete Dry Run

Input

75824

Initially

```
count = 0
```

Iteration 1

```
count = 1
```

```
n = 7582
```

Iteration 2

```
count = 2
```

```
n = 758
```

Iteration 3

```
count = 3
```

```
n = 75
```

Iteration 4

```
count = 4
```

```
n = 7
```

Iteration 5

```
count = 5
```

```
n = 0
```

Loop Stop.

Answer

5

Dry Run (Negative Number)

Input

-987

Step 1

`Math.abs(-987)`

↓

987

Loop

987

↓

98

↓

9

↓

0

Count

3

Answer

3

Time Complexity

Har iteration me ek digit remove hoti hai.

Agar number me d digits hain, to loop d baar chalega.

Aur

$$d = \log_{10}(N) + 1$$

Isliye

$$\text{Time Complexity} = O(\log N)$$

Proof

Example

`N = 123456`

Digits

6

Loop bhi

6

baar hi chalega.

Space Complexity

Extra variables

- `count`
- `n`

Bas.

Input size ke saath memory nahi badhti.

`Space Complexity = O(1)`

Advantages

- Interview standard solution.
 - Easy to understand.
 - Extra memory nahi lagti.
 - Sabhi integer values ke liye kaam karta hai (0 aur negative ko handle karne ke baad).
-

Disadvantages

- Har digit process karni padti hai.
 - Bahut bade numbers ke liye logarithm formula se thoda slow hai (although Big-O same rehta hai).
-

Common Mistakes

Mistake 1

```
while(n > 0)
```

Ye negative input ke liye kaam nahi karega.

Correct:

```
n = Math.abs(n);
```

Ya negative ko alag handle karo.

Mistake 2

0 ka special case bhool jana.

Example

Input

0

Loop chalega hi nahi.

Output

0

☐ Wrong

Correct

1

Mistake 3

```
count = count + 10;
```

☐ Wrong

Correct

```
count++;
```

Edge Cases

Input

0

Output

1

Input

5

Output

1

Input

1000

Output

4

Input

-45678

Output

5

Interview Me Kaunsi Approach Likhen?

□ Approach 1 (Division by 10)

Kyun?

- $O(\log N)$
 - $O(1)$ Space
 - Simple
 - Interview Standard
 - Sabse zyada use hone wali approach
-

Part 2 me hum cover karenge:

1. **Approach 2 – String Method**
2. **Approach 3 – Logarithm Formula ($\log_{10}$)**
3. $\log_{10}(0)$ ka edge case
4. Kab String use karna chahiye aur kab avoid karna chahiye
5. Interview me kaunsi approach best hai aur kyun

next

Day 13 – Count Digits in a Number (Part 2)

Aaj hum baaki 2 **interview-relevant approaches** cover karenge.

- □ Approach 2 – String Method
 - □ Approach 3 – Logarithm ($\log_{10}$) Method
-

Approach 2 – String Method

Idea

Number ko String me convert karo.

String ki length hi digits ki count hogi.

Example:

12345

↓

"12345"

↓

Length = 5

Bas.

Logic

Input

987654

Convert

"987654"

Length

6

Answer

6

Java Code

```
import java.util.Scanner;

public class CountDigitsString {

    public static int countDigits(int n) {

        String number = Integer.toString(Math.abs(n));

        return number.length();

    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number : ");

        int n = sc.nextInt();

        System.out.println("Digits = " + countDigits(n));

    }

}
```

Line by Line Explanation

Line

`Math.abs(n)`

Example

-1234

↓

1234

Minus sign digit nahi hota.

Line

`Integer.toString(...)`

Example

12345

↓

"12345"

Line

`number.length()`

String ke total characters return karta hai.

"12345"

↓

5

Dry Run

Input

-7865

Step 1

`Math.abs()`

↓

7865

Step 2

`Integer.toString()`

↓

"7865"

Step 3

`length()`

↓

4

Answer

4

Time Complexity

Conversion

$O(d)$

Length

$O(1)$

Overall

$O(\log N)$

Space Complexity

Ek String ban rahi hai.

Space Complexity = $O(\log N)$

Advantages

- Code bahut chhota.
 - Samajhne me easy.
 - Beginners ke liye achha.
-

Disadvantages

- Extra memory use hoti hai.
 - Interview me mathematical approach zyada preferred hoti hai.
-

Approach 3 – Logarithm Method ☐ ☐ ☐ ☐

Ye ek mathematical shortcut hai.

Main Formula

Agar

$N > 0$

to

$\text{Digits} = \text{floor}(\log_{10}(N)) + 1$

Ye formula direct answer de deta hai.

Example 1

Input

12345

Calculation

$\log_{10}(12345)$

≈ 4.091

Floor

4

Add

$4 + 1 = 5$

Answer

5

Example 2

Input

999

$\log_{10}(999)$

≈ 2.99

Floor

2

Add

3

Correct.

Java Code

```
import java.util.Scanner;

public class CountDigitsLog {

    public static int countDigits(int n) {

        if (n == 0) {
            return 1;
        }

        n = Math.abs(n);

        return (int) Math.log10(n) + 1;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number : ");

        int n = sc.nextInt();

        System.out.println("Digits = " + countDigits(n));
    }
}
```

Line by Line Explanation

Line

```
if (n==0)
```

Sabse important line.

Kyun?

Kyunki

```
log10(0)
```

exist hi nahi karta.

Java me

```
Math.log10(0)
```

↓

```
-Infinity
```

Milta hai.

Isliye 0 ko alag handle karna zaruri hai.

Line

```
Math.abs(n)
```

Negative number ko positive banata hai.

Line

```
Math.log10(n)
```

Example

```
1000
```

↓

```
3
```

Line

```
(int) Math.log10(n)+1
```

Example

98765

↓

4.99

Integer casting

4

Add

5

Answer

5

Dry Run

Input

54321

Calculation

$\log_{10}(54321)$

≈ 4.734

Floor

4

Add

5

Answer

5

Time Complexity

`Math.log10()` ek built-in mathematical operation hai.

Interview aur DSA me ise **constant time** maana jata hai.

Time Complexity = $O(1)$

Note: Ye theoretical analysis hai. Internally library ka implementation alag ho sakta hai, lekin algorithmic analysis me $O(1)$ hi maana jata hai.

Space Complexity

Koi extra data structure nahi.

Space Complexity = $O(1)$

Advantages

- Sabse chhota code.
 - Constant time ($O(1)$) maana jata hai.
 - Mathematical approach.
-

Disadvantages

- 0 ko alag handle karna padta hai.
 - Floating-point calculations hoti hain.
 - Bahut bade numbers ya floating precision ke concepts beginners ke liye confusing ho sakte hain.
 - Interviews me aksar loop-based approach ko preference di jati hai kyunki usse logic clearly dikhai deta hai.
-

Common Mistakes

Mistake 1

```
Math.log10(0)
```

☐ Invalid

Hamesha pehle:

```
if (n==0)
```

check karo.

Mistake 2

```
Math.log(n)
```

☐ Wrong

Ye natural logarithm hai.

Correct:

```
Math.log10(n)
```

Mistake 3

Negative number

```
-123
```

par direct

```
Math.log10(n)
```

lagana.

☐ Wrong

Pehle

```
Math.abs(n)
```

use karo.

Comparison of All Approaches

Feature	Division	String	Logarithm
Time	$O(\log N)$	$O(\log N)$	$O(1)$
Space	$O(1)$	$O(\log N)$	$O(1)$
Extra Memory	☐ No	☐ Yes	☐ No
Interview Preferred	☐ ☐ ☐ ☐ ☐	☐ ☐	☐ ☐ ☐
Beginner Friendly	☐	☐	Medium

Interview Me Kaunsi Approach Likhen?

Normal Coding Interview

☐ Approach 1 (Division by 10)

Kyun?

- Sabse common.
- Logic clearly dikhata hai.
- $O(1)$ Space.
- Kisi built-in function par depend nahi.

Mathematical Discussion ya Competitive Programming

☐ Approach 3 (Logarithm)

Agar interviewer specifically mathematical solution puche.

String Approach

Sirf alternative solution ke roop me mention karo, primary solution ke roop me nahi.